CRF State Summary v20

Session: Clifford Symmetry, Grade Current, Signed K35, Born Rule Resolved

Ougit Jarittum ORCID: 0009-0009-4451-6811

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Session summary. Starting from v19, this session investigated the Clifford algebra connection and resolved a key open question.
Four documents produced: CRF Paper v7 (Form Consistency, Space–Matter Split, $\beta = d_s\mathcal{G}$), CRF Clifford (11 proved connections), CRF Clifford 3 (Born rule resolved, grade current, 4th f_{II} interpretation), CRF StateSummary v20.
Three major results: (1) Born rule $\neq$ Clifford product (proved by linearity). (2) Signed K35 identity: $S^- = -n_{\mathrm{m}}(n_{\mathrm{m}}+1) = -30$ (exact). (3) $f_{\mathrm{II,crit}}$ acquires a fourth interpretation as critical Clifford charge density.

Key Numbers (unchanged)

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.94003$	$\kappa = 0.15007$	$K^* = 0.11750$
$n = 8$	$d_s = 3$	$\beta = 2.2121$	$F = 1.5377$	$T_{\mathrm{avg}} = 35.57$
$\varphi = 1.61803$	$e = 2.71828$	$\pi = 3.14159$	$m = \varphi$	$n_{\mathrm{m}} = 5$
$\mathcal{G} = 0.73737$	$f_{\mathrm{II,crit}} = 0.14700$	$n = d_s + n_{\mathrm{m}}$		

New Results This Session

Signed K35 identity (algebraically exact)

K35 pair

Unsigned (K35): $S^+ = \sum_k \binom{d_s}{k} k(n-k) = n(n+1) = 72$

Signed (new): $S^- = \sum_k \binom{d_s}{k} k(n-k) \cdot \mathrm{sign}(e_k^2) = -n_{\mathrm{m}}(n_{\mathrm{m}}+1) = -30$

where $\mathrm{sign}(e_k^2) = +1$ for grades 0,1 and -1 for grades 2,3.

From the pair: space contribution $= (S^+ + S^-)/2 = 21 = d_s(n-1)$; matter contribution $= (S^+ - S^-)/2 = 51 = 17n/8$; ratio $= 17/7$ (exact rational).

Born rule is not Clifford multiplication (proved)

Proof by linearity: Born rule $E[P_{\text{new}}] = P_{\text{avg}}$ is linear; Clifford product is bilinear. They cannot be equal for all states.

Correct framing: Cl(3) is the *symmetry group* of the CRF mode space (grade-preserving, static structure), not the operation of the Born rule (linear stochastic dynamics).

Grade current equation

The Clifford charge $C(P_i)$ satisfies:

$$\frac{dC_i}{dt} = \sum_j W_{ij}(C_j - C_i) + S_i(t)$$

where $S_i = +1$ at crystallisation events. Total charge $Q = \sum_i C_i$ is conserved by the diffusion term alone.

Fourth interpretation of $f_{\text{II,crit}}$

#	Name	Reading of $f_{\text{II,crit}} = 0.14700$
1	Percolation	Critical fraction for 3D vector-Ω subgraph to span
2	Crossing prob.	$P(N \geq 1 \,\vert\, \text{Poisson}(\sqrt{\varphi}/n))$
3	Phase II frac.	Steady-state fraction of crystallised nodes (V20.3)
4	Cl(3) charge	Critical Clifford charge density for d_s-lock

Prediction: 5 Psi-clusters = 5 grade sectors

Grade sectors of Cl(3) not used by space (grade-1):

- Outer clusters ($\pm 2\Delta_\Psi$): grades 0 and 3 (SO(3) singlets, $|\Omega| \approx 0$ or large)
- Inner clusters ($0, \pm\Delta_\Psi$): grade-2 triplet, each with Ω in one coordinate plane

Testable: extract per-cluster Ω distributions from V18 data.

Document Status

Document	Version	Key content
CRF Paper v7	**new**	Form theorem, split, $\beta = d_s\mathcal{G}$
CRF Clifford	**new**	11 proved connections, signed K35
CRF Clifford 3	**new**	Born rule resolved, grade current
CRF FormStructure	current	K35 uniqueness, proves Axiom 0
CRF MatterSplit	current	Powerset split, Hodge duality
CRF StateSummary	**v20**	This document

Open Items

1. **Simulation test: 5 clusters = 5 grade sectors.** Run V18 or equivalent; extract Ω per Psi-cluster. Check whether grade-2 clusters have Ω planar structure.
2. **Dispersion relation for grade current.** The grade current equation gives diffusive transport. Is there a drift term? If so, what is the wave speed?
3. **Collective Cl(3) structure of N nodes.** Each node is a Cl(3) element. What is the algebra of the full N-node system?
4. $\alpha\sqrt{\varphi} \approx D_0^2$ **(gap 0.22%).** If exact, $\varepsilon_0 = \alpha/(72\sqrt{\varphi})$ and CRF has zero free parameters. Current precision insufficient; needs tighter bound or proof.
5. **P0-H Subject 4 (bracketed).** Flat 50% prediction. $n \geq 50$ trials.

Transfer to next session

$\beta = d_s\mathcal{G}$ (gap 0.0006%). $n = d_s + n_\mathrm{m} = 3+5$ (exact). K35: $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$ (proved unique by form theorem). $\varepsilon_0 \leftrightarrow w_\mathrm{DE}$ (DESI gap 0.03%).

New (v20):
Signed K35: $S^- = -n_\mathrm{m}(n_\mathrm{m}+1) = -30$ (exact pair with K35).
Born rule $\neq$ Clifford product (proved by linearity).
Cl(3) = symmetry group, not dynamics.
Grade current: $dC_i/dt = \sum_j W_{ij}(C_j - C_i) + S_i$ (reaction-diffusion).
$f_{\mathrm{II,crit}}$ = critical Clifford charge density (4th interpretation).
Prediction: 5 Psi-clusters = 5 grade sectors (testable via simulation).

The universe lives within Cl(3), constrained by its grade structure.

Space is grade-1. Matter is grades 0, 2, 3.
The Born rule moves charge among grades.
Crystallisation creates it.

 Ougit Jarittum · ORCID: 0009-0009-4451-6811